

Web Accessibility

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

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Administrivia

- HW3 due Wednesday 11am: Try Out Assistive Tech
- Guest Speaker Peya Mowar this Wednesday
 - Making PDFs Accessible with AI

Accessibility at CMU

- Come to the Accessibility Lunch Series today!
 - Mondays 12:30-1:30pm NSH 3305
 - Free pizza!
 - Valerie Sazonov (University of Alabama) –
“Computer Use Agents Operating Assistive Technologies”
- Sign up for the accessibility at CMU mailing list!
 - <https://accessibility.cs.cmu.edu>

Web Accessibility Evaluation Tools

- WAVE: <https://wave.webaim.org/extension/>
- Axe: <https://www.deque.com/axe/>
- Accessibility Insights: <https://accessibilityinsights.io>

WAVE Demo

- Install WAVE as an extension to Chrome, Firefox, or Edge
- Let's evaluate some local web sites.
 - <https://duquesneincline.org/>

Participation Activity

- Pair up with a neighbor.
- Browse to <https://gatewayclipper.com>.
- Run the WAVE web accessibility checker.
- Pick two reported errors or alerts (each should be a different kind)
- Write a post with 3 parts to #in-class and tag your partner.
 1. What/where is the error?
 2. What's actually wrong with it?
 3. How would you change the HTML to fix it?

Participation Activity

- Do this activity by yourself.
- Run WAVE on the 10 most common web sites you personally visit on a daily or weekly basis.
- Post to #in-class:
 - What are the most common errors you found?
 - Are these real errors? Or, is WAVE being overly dramatic?
- Did you find a website with no errors at all?
 - If you can find one with no errors, you'll get a prize!

Alt-Text

- Text description of an image
- Alt-text must be informative and descriptive of picture content.
- It can be funny, but this is not the place to add easter eggs or jokes.

Qualities of good alt-text

1. What kind of image is it? (hand-drawn, polaroid, infographic, screenshot, chart, map, diagram, photo, etc)
2. What is the most salient aspect of the image?
3. Is the background important to place the image in context?

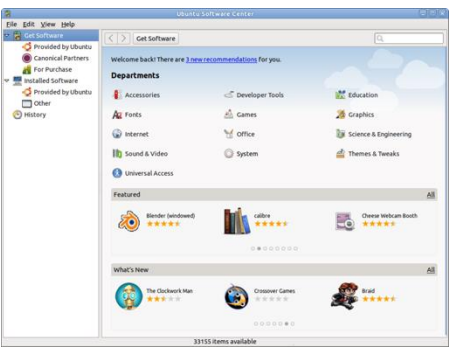
Image Type



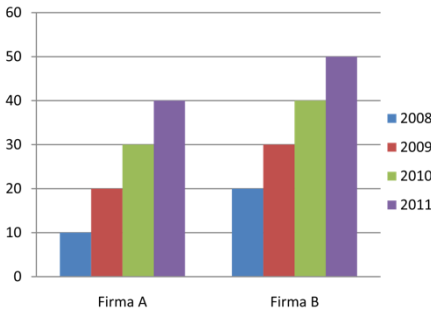
Handdrawn



Polaroid



Screenshot



Chart



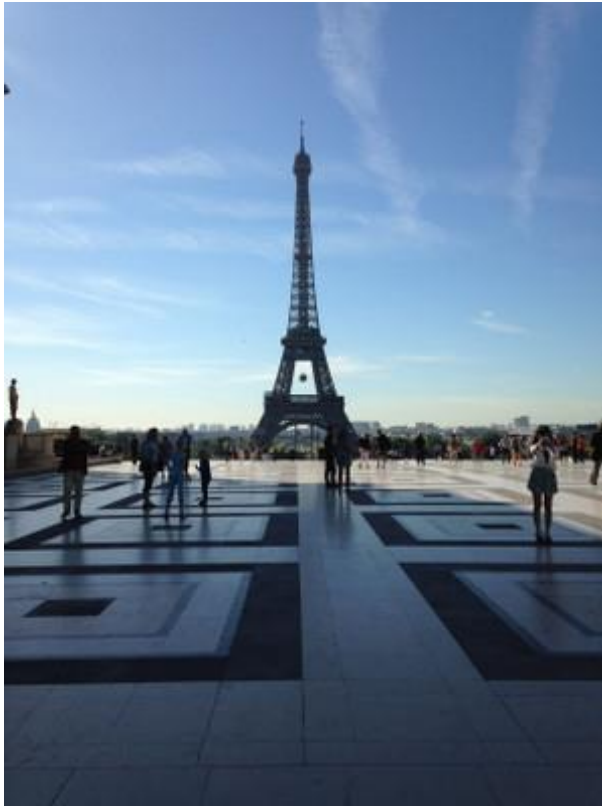
Map

Infographic



Photo

Most Salient Feature



Importance of Background Context



Participation Activity

- Pair up with a neighbor.
- Write a visual description of your neighbor as they appear to you today for the benefit of a blind visitor.
- Post your visual description in **#in-class** and tag the post with your neighbor.
- What was important enough about your neighbor to describe?

Participation Activity

- Stay with your partner.
- Take out your phones and take a selfie.
- Write the alt-text description for your selfie and post it and the picture in **#in-class** as a reply to the visual description of yourself.
- How different is this alt-text description than the visual description your partner wrote about you?

Participation Activity

- Open up ChatGPT or your favorite LLM.
- Copy and paste your selfie into the LLM and ask it to write an alt-text description for the picture.
- Post the LLM-generated alt-text to **#in-class** as a reply to your selfie post.
- How does the AI's alt-text differ from the one you wrote?
- Why?

Alt-text Discussion

- Who is best at writing alt-text?
- Does the purpose of the picture matter?
- Should it?

WCAG Guidelines - Perceivable

Distinguishability

- Color
- Contrast
- Size (Text/Image)
- Audio (Volume/Noise)
- Visual Presentation
- Reflow

Media

- Alternatives to Audio?
- Alternatives to Video?
- Alternatives to Text?
- Captions
- Audio Descriptions

WCAG Guidelines - Operable

Input

- Keyboard accessible
- Timing Adjustable
- Interruptibility
- Pause/Stop/Hide
- Timeouts

Navigable

- Focus Order and Appearance
- Headings and Labels
- Location (Breadcrumbs)
- Link Purpose

WCAG Guidelines - Understandable

Readable

- Reading Level
- Unusual Words
- Abbreviations
- Pronunciation

Predictable

- Focusing on an element
- Consistent navigation
- Consistent help

WCAG Guidelines - Robust

- Compatible
 - Name, Role, Value
 - Status Messages



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Witness in fatal ICE shooting disputes the government's self-defense claim in exclusive interview

Breaking News

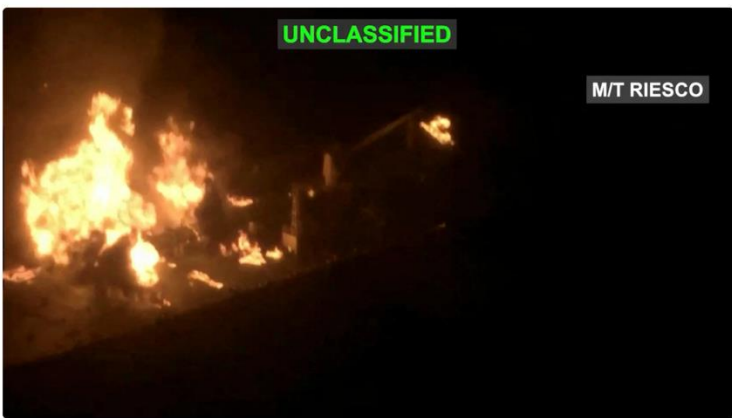


Democratic Rep. Chris Pappas will face GOP former Sen. John E. Sununu in New Hampshire Senate race, CNN projects

Fact Check



Iran launches missile barrage at Jordan after US strikes 5 Iranian tankers



UNCLASSIFIED

M/T RIESCO

US Central Command/Handout/Reuters

US forces struck the vessels after Iran attempted to attack a US Navy ship, CENTCOM says

US targets Iran's airline industry as part of pressure campaign

Scores wounded as Iran-backed Houthis attack Saudi Arabia in major regional escalation

WASHINGTON



Catch up on today's headlines



Tiafoe gives Michelsen an emotional embrace after his US Open victory



NCAA says it won't overturn Michigan's controversial win after MAC raised replay concerns

STREAMING NOW



Participation Activity

- Pair up with a new neighbor
- Evaluate cnn.com with respect to WCAG POUR attributes
- For P, O, U, and R, identify one web element on cnn.com that obeys the attribute.
 - If you can, find another one that fails.
- Post descriptions of the four web elements to **#in-class**, paired with the specific WCAG POUR issue that succeeded or failed.
- Tag your partner in your post.